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MILITARY SPECIFICATION

SWITCH, SQUIB
MARK 65 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Squib Switch Mark 65 Mod 0. (See 6.1 for intended use and description).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein except as noted.

SPECIFICATION

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-vapor Proof or Waterproof, Flexible

FSC - 1375

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibration of Shipboard Equipment

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 497832	Switch, Squib, Mark 65 Mod 0 and all documents listed thereon

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications

Interstate Commerce Commission

49 CFR Part 71 to 78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations and are available from the Superintendent of Documents, Government Printing Office, Washington D. C. 20402. Orders for the above publication should cite "the latest issue and supplements thereto".)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the switch shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 497832.

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3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the switch shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Safety. The supplier or manufacturer, as applicable, shall provide adequate safety precautions for protection of personnel during handling of the switch (See 6.5.)

3.4 Performance and product characteristics. The switch shall meet the following performance and product characteristics.

3.4.1 Circuit requirements, before functioning.

3.4.1.1 Contact rating, circuits 1 and 2. The switch shall be capable of carrying a current of 5 amperes continuously through circuits 1 and 2. See Figure 1 for circuit identification.

3.4.1.2 Resistance, circuits 1 and 2. The normally closed circuits of the switch, circuits 1 and 2, shall have a resistance of not greater than 0.020 ohm.

3.4.1.3 Resistance, initiation circuit. The initiation circuit of the switch shall have a resistance of 5.2 plus or minus 1.2 ohms. See Figure 1 for circuit identification.

3.4.1.4 Insulation resistance. After being conditioned at a temperature of 75 plus or minus 5 degrees Fahrenheit (F) and a relative humidity of 75 plus or minus 5 percent for a period of not less than twenty-four hours, the insulation resistance shall not be less than 100 megohms. When the insulation resistance is measured during or subsequent to the environmental tests as specified in 4.5, there shall be no temperature or humidity conditioning other than that required by the individual environmental test and the insulation resistance shall then be 20 megohms minimum. The test potential shall be 500 plus or minus 50 volts DC. The insulation resistance shall be measured between the following mutually insulated points:

a. The case

b. Either terminal of circuit 1

c. Either terminal of circuit 2

d. One terminal of circuit 3

e. Other terminal of circuit 3

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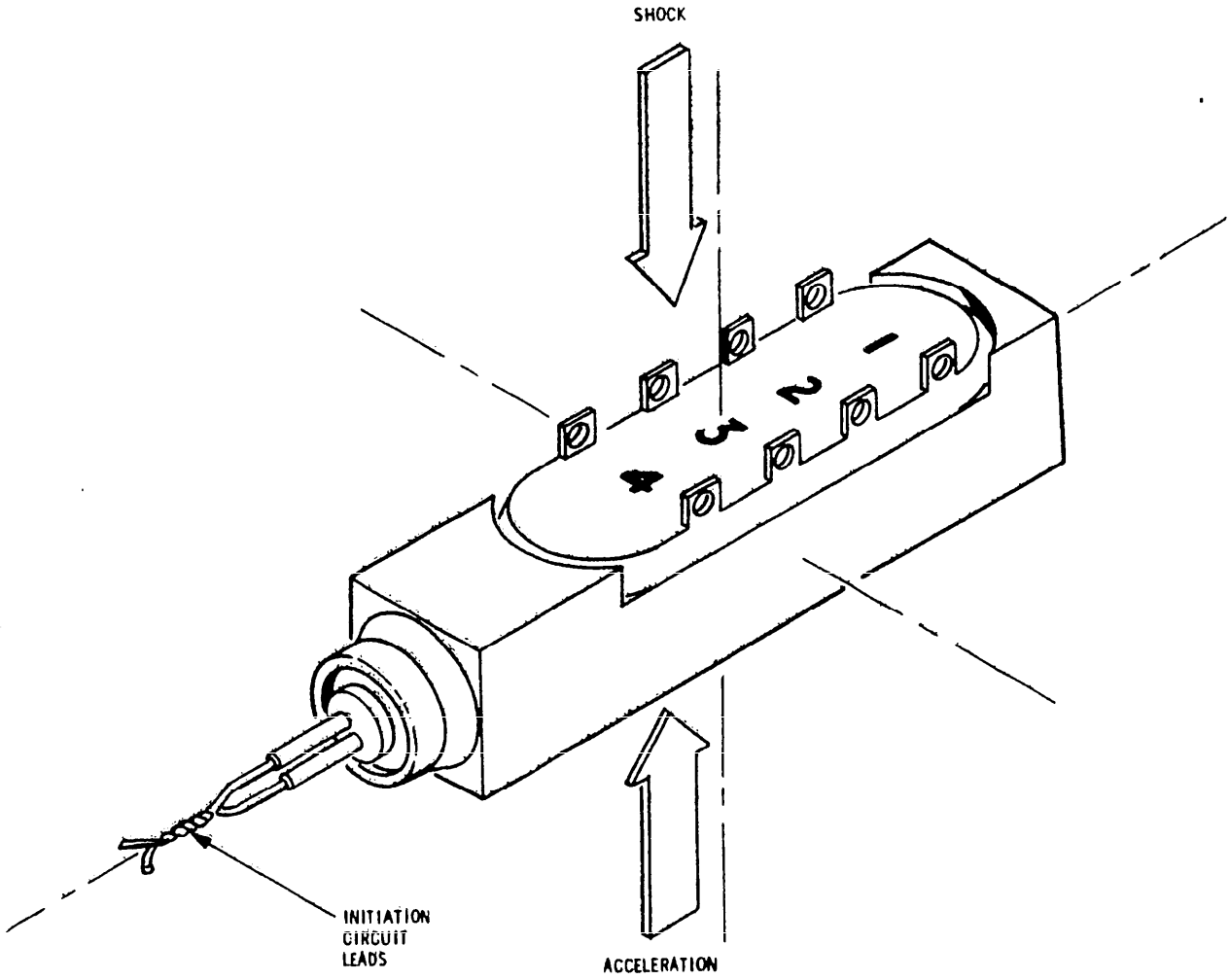


Figure 1. Environmental Testing Axes and Switch Circuits

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f. One terminal of circuit 4

g. Other terminal of circuit 4

3.4.2 Electrical sensitivity, initiation circuit. The initiation circuit of the switch shall be capable of conducting a 10 milliampere direct current for not less than 5 minutes without igniting or deteriorating the explosive driver.

3.4.3 Function. The switch shall not rupture or fragment when functioned.

3.4.3.1 Functioning pulse. The switch shall open circuits 1 and 2 and close circuits 3 and 4, when a 7000 erg pulse is dissipated through the initiation circuit within 1 millisecond.

3.4.3.2 Functioning time. The switch shall open circuits 1 and 2 and close circuits 3 and 4 in not greater than 10 milliseconds from the moment the functioning pulse is applied.

3.4.4 Circuit requirements, after functioning.

3.4.4.1 Contact rating, circuits 3 and 4. The switch shall be capable of carrying a direct current of 5 amperes continuously through circuits 3 and 4. See Figure 1 for circuit identification.

3.4.4.2 Resistance, circuits 3 and 4. The resistance between the terminals of circuits 3 and 4 shall not be greater than 0.020 ohm.

3.4.4.3 Insulation resistance. After being conditioned at a temperature of 75 plus or minus 5 degrees Fahrenheit (F) and a relative humidity of 75 plus or minus 5 percent for a period of not less than twenty-four hours, the insulation resistance shall not be less than 100 megohms. When the insulation resistance is measured during or subsequent to the environmental tests as specified in 4.5, there shall be no temperature or humidity conditioning other than that required by the individual environmental test and the insulation resistance shall then be 20 megohms minimum The test potential shall be 500 plus or minus 50 volts DC. The insulation resistance shall be measured between the following mutually insulated points:

a. The case

b. One terminal of circuit 1

c. Other terminal of circuit 1

d. One terminal of circuit 2

e. Other terminal of circuit 2

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f. Either terminal of circuit 3

g. Either terminal of circuit 4

3.5 Environmental requirements. Unless otherwise specified, the switch shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.5.1 Temperature, storage condition. The switch shall withstand temperatures from minus 65 to plus 160 degrees F.

3.5.2 Vibration.

3.5.2.1 Standby condition. The switch shall withstand vibration as specified for Type I equipment in MIL-STD-167 up to 25 cycles per second (cps).

3.5.2.2 Operating condition. The switch shall withstand vibration through a frequency range of 20 to 2000 cps at 0.10 inch double amplitude or 10g (g = unit of gravitational acceleration) whichever is less.

3.5.3 Acceleration. The switch shall withstand acceleration of 25g for 4 seconds minimum applied in the direction shown in Figure 1.

3.5.4 Shock.

3.5.4.1 Standby condition. The switch shall withstand shocks of 40g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 5 milliseconds.

3.5.4.2 Operating condition.

3.5.4.2.1 Major. The switch shall withstand a shock of 2400g peak magnitude in the direction shown in Figure 1. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 millisecond.

3.5.4.2.2 Minor. The switch shall withstand two shocks of 1200g peak magnitude 90 degrees apart and perpendicular to the major shock axis. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 millisecond.

3.5.5 Temperature, operating condition. The switch shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of

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this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the switch shall be classified as follows:

a. Preproduction and periodic production inspection. (See 4.5.)

b. Quality conformance inspection. (See 4.6.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order the number of products in each lot shall be two thousand (2000).

4.3.2 Preproduction sample. A preproduction sample of fifty (50) switches, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switch by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.3 Periodic production sample. From each succeeding lot of two thousand (2000) or less, a random sample of fifty (50) switches shall be selected. These sample units shall be subjected to the tests of 4.5. The switches shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5. (See Table I.) When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a. Determine the cause of the defect.

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b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the switch inspected as specified in 4.6.

Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Quality Conformance Tests	4.6*	50	50
Temperature, Storage Condition	4.5.1.1	50	50
Quality Conformance Tests	4.6	12**	12**
Vibration, Standby Condition	4.5.1.3	38	38
Shock, Standby Condition	4.5.1.5	38	38
Quality Conformance Tests	4.6	12**	12**
Vibration, Operating Condition	4.5.1.4	26	26
Acceleration	4.5.1.2	26	26
Shock, Operating Condition	4.5.1.6	26	26
Quality Conformance Tests	4.6	12**	12**
Temperature, Operating	4.5.1.7		
Low Extreme	4.5.1.7.1	7	7
Quality Conformance Tests	4.6	7**	7**
High Extreme	4.5.1.7.2	7	7
Quality Conformance Tests	4.6	7**	7**

*Except for functioning tests, 4.6.2.3, and circuit tests, 4.6.2.4.
**Samples are expended during functioning.

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4.4 Test equipment.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests.

Equipment	Capability
a. DC power supply, current regulated	Regulate direct current at 5.0 plus 0.0 minus 0.5 ampere through test item load (0.10 ohm maximum)
b. Voltmeter	Measure 0.00 to 0.30 volt DC; accuracy plus or minus 3 percent
c. Low resistance measuring device	Measure 0.00 to 0.05 ohm; accuracy plus or minus 5 percent
d. Ohmmeter, current limited	Measure 0.0 to 10.0 ohms; accuracy plus or minus 5 percent; maximum current 5 milliamperes
e. Conditioning chamber	Maintain temperature at 75 plus or minus 5 degrees F and a relative humidity of 75 plus or minus 5 percent
f. Megohmmeter	Measure 0.0 to 100 megohms; accuracy 10 percent at 20 and 100 megohms; 500 plus or minus 50 volts DC potential
g. DC power supply current regulated	Regulate direct current at 10 plus 0 minus 1 milliamperes through test item initiation circuit (approximately 5.2 ohms)
h. Capacitor	40 microfarad plus or minus 10 percent capable of being charged to 6.5 plus or minus 0.25 volts DC
i. Time interval counter	Measure intervals from 0.0 to 10.0 milliseconds; accuracy plus or minus 10 percent.

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs, are required to perform the preproduction and periodic production tests:

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Equipment	Capability
a. Temperature chamber	Minus 65 plus 5 minus 0 degrees F Plus 35 plus 5 minus 0 degrees F Plus 120 plus 0 minus 5 degrees F Plus 160 plus 0 minus 5 degrees F
b. Vibration testing machine	Control frequency from 20 cps to 2000 cps while subjecting the test item to 0.10 inch double amplitude or 10g acceleration, whichever is less: maintain direction of vibration along specified axes
c. Centrifuge	0 to 27.5g: maintain acceleration at 25 plus 2.5 minus 0g
d. Shock equipment, standby	Apply shocks of 40 plus or minus 4g: a one-half sine wave pulse shape having a duration of 5 plus 3 minus 0 milliseconds; maintain direction along specified axes
e. Shock equipment, operating	Apply shocks of up to 2400g plus or minus 10 percent: a one-half sine wave pulse shape having a time duration of 1 plus or minus 0.25 millisecond: maintain direction along specified axes

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples shall be subjected to the tests specified in 4.5.1 after satisfactorily passing the quality conformance inspections of 4.6.2.1. 4.6.2.2 and 4.6.2.5

4.5.1 Test procedure. Subject the switch to the test sequence listed in Table I. Unless otherwise specified in the individual test paragraph, the quality conformance inspection, as specified in Table I shall be performed subsequent to environmental tests under test conditions specified in 4.6.1. Failure of the switch to pass any one of these tests shall constitute a defective unit.

4.5.1.1 Temperature, storage condition. The switch shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

4.5.1.2 Acceleration. The switch, mounted as shown in Figure 2, shall be subjected to acceleration of 25 plus 2.5 minus 0g in the direction shown in Figure 1 for not less than 4 seconds.

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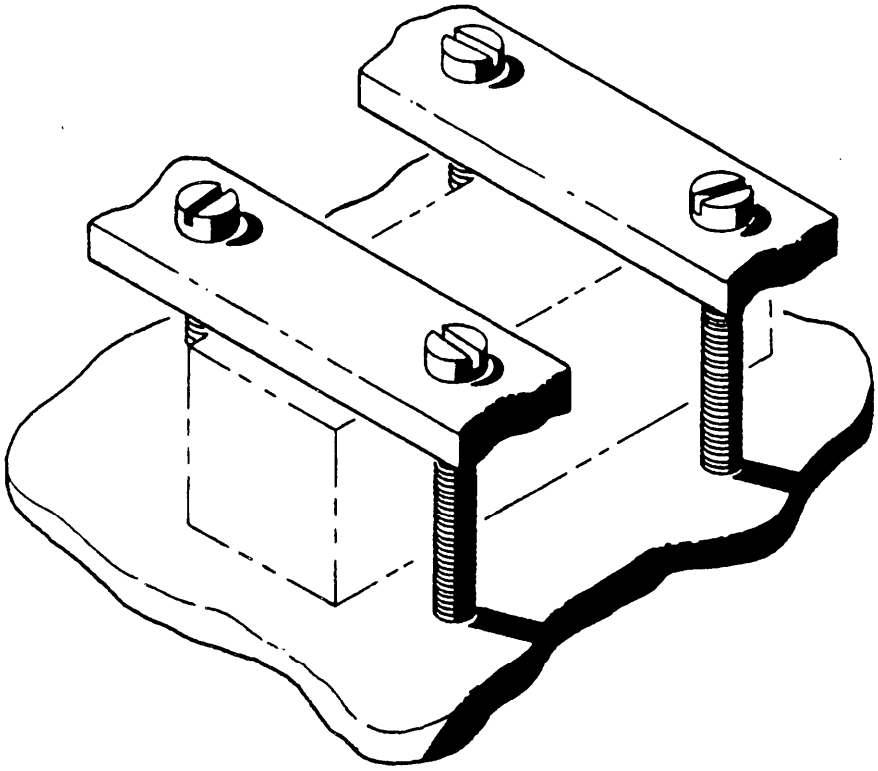


Figure 2. Holding Fixture

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4.5.1.3 Vibration, standby condition. The switch, rigidly mounted as shown in Figure 2, shall be subjected to the vibration tests for Type I equipment as specified in MIL-STD-167 up to 25 cps in each of the three axes shown in Figure 1.

4.5.1.4 Vibration, operating condition. The switch, rigidly mounted as shown in Figure 2, shall be subjected to vibration from 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less, along each of three mutually perpendicular axes. The frequency shall be increased uniformly from 20 to 2000 cps. The frequency shall be traversed in not less than 2.5 minutes, once in each axis.

NOTE: Instrumentation used for the shock test shall not employ a filter with a cut-off frequency below 10 kc.

4.5.1.5 Shock standby condition. Subject the switch to shocks of 40 plus or minus 4g peak magnitude with a pulse shape approximating a one-half sine wave, having a duration of 5 plus 3 minus 0 milliseconds. Shocks shall be applied three times in each direction along each of three mutually perpendicular axes as shown in Figure 1. Mount the switch in a manner similar to that shown in Figure 2.

4.5.1.6 Shock, operating condition. The switch shall be mounted as shown in Figure 2, and subjected to the following tests:

4.5.1.6.1 Major. Subject the switch to a shock of 2400 plus or minus 240g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shock in the direction shown in Figure 1.

4.5.1.6.2 Minor. Subject the switch to two shocks of 1200 plus or minus 120g peak magnitude. The shocks shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shocks in two directions 90 degrees apart and perpendicular to the major shock axis.

4.5.1.7 Temperature, operating condition. Divide the sample into two equal parts as indicated in Table I. One-half of the sample shall be subjected to the low temperature test as specified in 4.5.1.7.1 and the remaining half of the sample shall be subjected to the high temperature test as specified in 4.5.1.7.2.

4.5.1.7.1 Low extreme. Subject the switch to a temperature of 35 plus 5 minus 0 degrees F for not less than 2 hours and while at that temperature, the switch shall pass the quality conformance inspection tests.

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4.5.1.7.2 High extreme. Subject the switch to a temperature of 120 plus 0 minus 5 degrees F for not less than 2 hours, and while at that temperature, the switch shall pass the quality conformance inspection tests.

4.6 Quality conformance inspection. Samples, selected using the plans and procedures set forth in Table II of this specification, shall be subjected to the tests and examinations as described in 4.6.2.

Table II. Quality Conformance Tests and Examinations and AQL

Test	Sampling Plan	AQL	Test Paragraph
Contact rating, circuits 1 and 2	*	0.25	4.6.2.1.1
Resistance, circuits 1 and 2	*	1.0	4.6.2.1.2
Resistance, initiation circuit	*	1.0	4.6.2.1.3
Insulation resistance	*	1.0	4.6.2.1.4
Electric initiation circuit sensitivity	*	0.25	4.6.2.2
Functioning pulse	*	0.25	4.6.2.3.1
Functioning time	*	1.00	4.6.2.3.2
Contact rating, circuits 3 and 4	*	0.25	4.6.2.4.1
Resistance, circuits 3 and 4	*	0.25	4.6.2.4.2
Insulation resistance	*	1.0	4.6.2.4.3
Visual examination	**	**	4.6.2.5
Preservation, packaging, packing, and marking	**	**	4.7

*MIL-STD-105D Inspection Level II, except that sample size shall not be less than fifty (50) switches for normal inspection regardless of lot size.
**One-hundred percent inspection: defect is cause for rejection of individual unit.

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4.6.1 Test conditions. Unless otherwise specified, the switch shall be subjected to the quality conformance inspection under the following conditions:

a. Temperature - Room ambient (65 to 95 degrees F)

b. Altitude - Normal ground atmospheric pressure

c. Humidity (Relative) - Room ambient up to 95 percent

WARNING

Provide a protective barrier for inspection personnel to prevent possible injury due to fragmentation during quality conformance inspection tests.

4.6.2 Test procedure. The switch shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the switch to pass the tests and examinations shall constitute a failure.

4.6.2.1 Circuit requirements before functioning.

4.6.2.1.1 Contact rating, circuits 1 and 2. Connect the voltmeter specified in 4.4.1b between the terminals of circuit 1 and, using the power supply specified in 4.4.1a, apply a current of 5 plus 0.0 minus 0.5 amperes between the terminals for 90 seconds. The voltage drop shall be not greater than 0.10 volt. Connect the voltmeter between the terminals of circuit 2. Apply a current of 5 plus 0.0 minus 0.5 amperes between the terminals for 90 seconds. The voltage drop shall be not greater than 0.10 volt.

4.6.2.1.2 Resistance, circuits 1 and 2. Using the low resistance measuring device specified in 4.4.1c measure the resistance between the terminals of circuit 1 and between the terminals of circuit 2. The resistance of either circuit shall not be greater than 0.020 ohm.

4.6.2.1.3 Resistance, initiation circuit. Using the ohmmeter specified in 4.4.1d measure the resistance between the leads of the initiation circuit. The resistance shall be 5.2 plus or minus 1.2 ohms.

WARNING

The electric initiation circuit must be twisted together during insulation resistance test to prevent inadvertent energizing of the switch.

4.6.2.1.4 Insulation resistance. Place the switch in a conditioning chamber and maintain the temperature at 75 plus or minus 5 degrees F and the relative humidity at 75 plus or minus 5 percent for a period of not less than

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twenty-four hours. Remove the switch from the chamber and within 10 minutes measure the insulation resistance between the mutually insulated points listed below. The applied potential shall be 500 plus or minus 50 volts DC. The insulation resistance shall not be less than 100 megohms. When this test is performed during or subsequent to the environmental tests specified in 4.5.1, there shall be no temperature or humidity conditioning other than that required by the individual environmental test; the insulation resistance shall then be 20 megohms minimum. The mutually insulated points are:

a. The case

b. Either terminal of circuit 1

c. Either terminal of circuit 2

d. One terminal of circuit 3

e. Other terminal of circuit 3

f. One terminal of circuit 4

g. Other terminal of circuit 4

4.6.2.2 Electric initiation circuit sensitivity. Using the power supply specified in 4.4.1g apply a direct current of 10 plus 0 minus 1 milliamperes through the electric initiation circuit for not less than 5 minutes. The switch shall not function.

4.6.2.3 Functioning tests. If any switch ruptures or fragments during functioning tests, the lot shall be rejected. Simultaneously perform tests of 4.6.2.3.1 and 4.6.2.3.2.

4.6.2.3.1 Functioning pulse. Subject the initiation circuit of the switch to the discharge from a 40 plus or minus 4 microfarad capacitor specified in 4.4.1h charged to 6.5 plus or minus 0.25 volts. The switch shall function.

4.6.2.3.2 Functioning time. Connect the time interval counter specified in 4.4.1i to the switch in such a manner that the application of the initiation pulse will start the time interval meter and the closing of the open circuits stop the meter. The time interval shall be not greater than 10 milliseconds.

4.6.2.4 Circuit tests, after functioning.

4.6.2.4.1 Contact rating, circuits 3 and 4. Connect the voltmeter specified in 4.4.1b between the terminals of circuit 3, and using the power supply specified in 4.4.1a, apply a current of 5 plus 0.0 minus 0.5 amperes between

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the terminals for 90 seconds. The voltage drop shall be not greater than 0.10 volt. Connect the voltmeter between the terminals of circuit 4. Apply a current of 5 plus 0.0 minus 0.5 amperes between the terminals for 90 seconds. The voltage drop shall be not greater than 0.10 volt.

4.6.2.4.2 Resistance, circuits 3 and 4. Using the instrument specified in 4.4.1c measure the resistance between the terminals of circuit 3 and between the terminals of circuit 4. The resistance of either circuit shall be not greater than 0.020 ohm.

4.6.2.4.3 Insulation resistance. Place the switch in a conditioning chamber and maintain the temperature at 75 plus or minus 5 degrees F and the relative humidity at 75 plus or minus 5 percent for a period of not less than twenty-four hours. Remove the switch from the chamber and within 10 minutes measure the insulation resistance between the mutually insulated points listed below. The applied potential shall be 500 plus or minus 50 volts DC. The insulation resistance shall not be less than 100 megohms. When this test is performed during or subsequent to the environmental tests specified in 4.5.1, there shall be no temperature or humidity conditioning other than that required by the individual environmental test: the insulation resistance shall then be 20 megohms minimum. The mutually insulated points are:

a. The case

b. One terminal of circuit 1

c. Other terminal of circuit 1

d. One terminal of circuit 2

e. Other terminal of circuit 2

f. Either terminal of circuit 3

g. Either terminal of circuit 4

4.6.2.5 Visual examination. Visual examination shall be made on each switch submitted for delivery to the procuring activity. The switch shall be examined for conformance to the requirements of good workmanship in 3.6. Nonconformance is cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The switch shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. (See 6.2b and 6.4.)

5.1.1 Level A.

5.1.1.1 Cleaning. The switch shall be cleaned in accordance with Method C-1, MIL-P-116.

5.1.1.2 Drying. The switch shall be dried in accordance with Method D-1, MIL-P-116.

5.1.1.3 Unit Packaging. The switch shall be individually packaged in accordance with Method IId of MIL-P-116.

5.1.2 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C as specified. (See 6.2c and 6.4.)

5.2.1 Level A. Unless otherwise specified in the contract, order or requisition, the switch shall be packed in exterior type shipping container, PPP-B-621, of uniform size and shape, with the minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to approximately 100 lbs. Containers shall be provided with a case liner, MIL-L-10547, and shall be closed and strapped in accordance with PPP-B-621.

5.2.2 Level C. The switch packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers will conform to the 49 CFR Part 71 to 78, or other carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and 49 CFR Part 71 to 78.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The switch is intended to be used to open or close electrical circuits in a depth charge fuze.

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6.1.2 Description. The switch is a gas actuated, four-pole, single-throw switch, consisting of two normally open circuits and two normally closed circuits. These circuits are closed and opened, respectively, when the initiation circuit is energized by an energy pulse of 7000 ergs dissipated over a period of one millisecond.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Required level of preservation and packaging

c. Required level of packing

d. Any exceptions to this specification.

6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Operating. The condition of the switch during missile operation.

6.3.1.2 Standby. Standby is the condition of the switch when aboard ship and is assembled with the end item.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

6.5 General safety precautions. The loading, assembly, and handling of explosives, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions as are pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provisions of the specification.

6.5.1 All loading operations should be conducted in a neat and orderly manner.

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6.5.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as milling, pouring, weighing, and charging, sifting, drying, pressing, casting, crimping, etc.

6.5.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which effect functioning should, in so far as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.4 The exposure of explosive material and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and while in process, in safety lockers or chests if in loading rooms, or in adequate ready or service magazines, located in accordance with intraplant distances when outside of loading rooms. For Navy explosvies loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives, shall apply

6.5.6 Proper care should be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed.

6.5.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacle. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures, and where necessary, employ suitable grounded conductive coverings for floors, work benches, and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices

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such as grounded bracelets for workers should be employed where operations are conducted with items which are usually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures; and low-energy type electric primers, detonators, and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency hearing apparatus, etc. Where necessary for safety, humidity (See 6.5.4) of work rooms should be increased, as required to lessen electrostatic effects without excessive moisture absorption.

6.5.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electronic soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators, or squibs.

6.5.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.5.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-S-82487(OS) SWITCH, SQUIB MARK 65 MOD 0

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

FORM 1426
1 APR 63

C-8978

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Naval Ordnance Systems Command
Washington, D.C. 20360

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